PROJECT PROCESS

Research, Planning and Preparation

- Investigated how quantum computing is commonly taught and identified the heavy use of mathematical notation as a barrier for beginners.
- Explored Qiskit and existing quantum simulators to understand how qubits and gates are represented computationally.
- Studied Bob Coecke's Quantum Picturalism and ZX-calculus, which inspired the visual approach used in the project.
- Created a six-week development plan covering research, coding, testing, refinement, and prototype completion.
- Selected Python for accessibility and sustainability, and chose Tkinter for simple, built-in graphical visualisation.

Development and Problem Solving

- Implemented quantum gate functions, including X, Z, H (Hadamard), and CNOT gates using Qiskit.
- Developed a custom Spider class and diagram engine on Tkinter to convert quantum circuits into ZX-style visual representations.
- Encountered technical challenges such as Qiskit installation issues and errors within the Spider class.
- Resolved these problems by verifying Python environments, troubleshooting module recognition, and implementing a correct __init__ method for storing spider properties such as qubit index and phase.

Testing, Evaluation and Final Outcome

- Conducted regular testing throughout development to verify both quantum simulations and visual outputs.
- Confirmed that gate behaviour, such as the Hadamard gate producing a 50/50 probability distribution, matched expected quantum results.
- Checked that circuits were accurately converted into ZX-style diagrams on Tkinter with correct spider connections and phase information.
- Gathered feedback from teachers and peers to assess clarity and ease of understanding.
- Refined the interface and visuals based on feedback, resulting in a working prototype that demonstrates how quantum concepts can be communicated through intuitive diagrams rather than complex mathematics.